SENTRY KEY IMMOBILIZER MODULE (Continued)

(9) Reinstall the steering column opening cover. (Refer to 23 - BODY/INSTRUMENT PANEL/STEERING COLUMN OPENING COVER - INSTALLATION).

(10) Reconnect the battery negative cable.

NOTE: If the SKIM has been replaced with a new unit, the Sentry Key Immobilizer System (SKIS) MUST be initialized before the vehicle can be operated. (Refer to 8 - ELECTRICAL/VEHICLE THEFT SECURITY - STANDARD PROCEDURE - SKIS INITIALIZATION).

TRANSMISSION CONTROL MODULE

DESCRIPTION

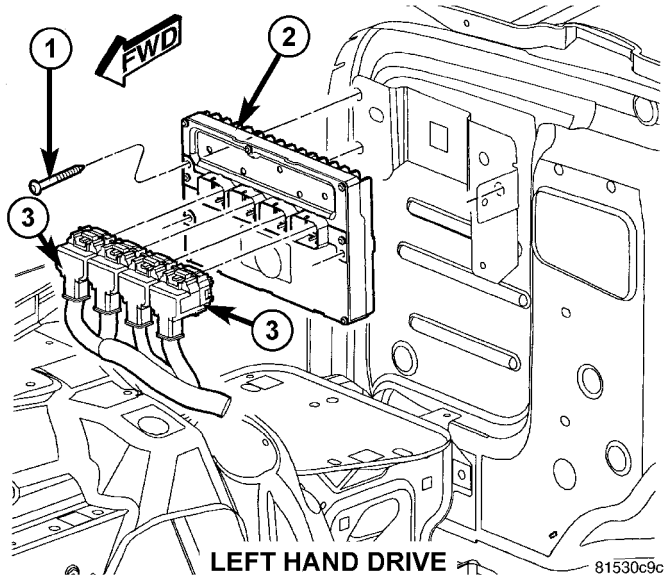

Fig. 12 PCM MOUNTING - LEFT HAND DRIVE

1 - MOUNTING SCREWS (3)
2 - PCM
3 - PCM ELECTRICAL CONNECTORS

The Transmission Control Module (TCM) is a submodule within the Powertrain Control Module (PCM) (2).

The PCM (2) is located on the right side of the dash panel in the engine compartment (Fig. 12) on left hand drive vehicles. The PCM is referred to as the NGC.

The PCM (1) is located on the left side of the dash panel in the engine compartment (Fig. 13) on right hand drive vehicles.

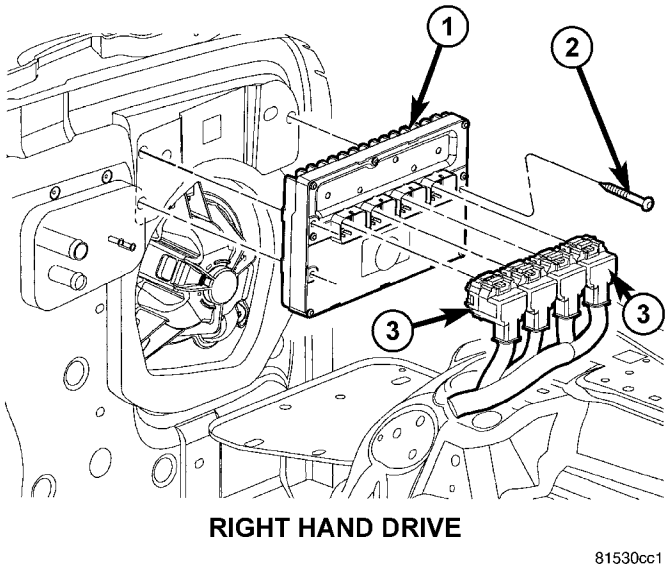

Fig. 13 PCM MOUNTING - RIGHT HAND DRIVE

1 - PCM
2 - MOUNTING SCREWS (3)
3 - PCM ELECTRICAL CONNECTORS

OPERATION

The TCM is the controlling unit for all electronic operations of the transmission. The TCM receives information regarding vehicle operation from both direct and indirect inputs, and selects the operational mode of the transmission. Direct inputs are hardwired to, and used specifically by the TCM. Indirect inputs originate from other components/modules, and are shared with the TCM via the PCI bus.

Some examples of **direct inputs** to the TCM are:

- Battery (B+) voltage
- Ignition “ON” voltage
- Transmission Control Relay (Switched B+)
- Throttle Position Sensor
- Crankshaft Position Sensor (CKP)
- Transmission Range Sensor (TRS)
- Pressure Switches (L/R, 2/4, OD)
- Transmission Temperature Sensor (Integral to TRS)
- Input Shaft Speed Sensor
- Output Shaft Speed Sensor

Some examples of **indirect inputs** to the TCM are:

- Engine/Body Identification
- Manifold Pressure
- Target Idle
- Torque Reduction Confirmation
- Speed Control ON/OFF Switch
- Engine Coolant Temperature
- Ambient/Battery Temperature
- Brake Switch Status
- DRB® III Communication